Lesson 3.2: Carbon Dioxide & Weather

Location: Classroom

Grade Level: High School

Time: Two class sessions of 50 minutes

Driving Questions

- What happens when there is more carbon dioxide in the atmosphere?
- How does increased carbon dioxide in the atmosphere affect weather patterns? How does this affect the ocean?

Lesson Synopsis

In this lesson, students investigate how rising atmospheric carbon levels influence weather patterns by trapping heat and driving changes to the Earth's climate. Building on their model from Lesson 3.1, students focus on understanding how human activities influence the greenhouse effect.

During the first day, students explore an initial model of how they think light energy from the sun interacts with the Earth and is released back out of the atmosphere in a normal setting. They then watch a video that describes the impact carbon dioxide can have on the atmospheric temperature and apply this information to their model to describe how human activities can increase the levels of carbon dioxide in the atmosphere.

During the second day, students further their understanding by applying their knowledge to a scenario to describe how a blanket traps heat and keeps an individual warm. They then expand on how carbon dioxide in the atmosphere acts like the "blanket" for Earth, trapping the heat and leading to an increase in global temperatures. Finally, students consider how these changes may ultimately impact the values their communities hold connected to Southern California's ocean, setting the stage for deeper exploration of particular ocean impacts in the next lesson.

What students will figure out:

- Human activities increase atmospheric carbon dioxide, which enhances the greenhouse effect by trapping more heat in Earth's atmosphere.
- Rising CO2 levels influence climate and weather patterns.
- Changes in climate impact local communities and ocean systems.

LEARNING OUTCOMES

By the end of this investigation, students will be able to…	You can assess this using…
1. Explain how an increase in atmospheric carbon traps heat energy on Earth.	Video during the *Navigate* section.

2. Draw a model showing how increased levels of carbon dioxide in the Earth's atmosphere act as a blanket and trap more heat.	Student models during the *Investigate* section.
3. Explain how rising atmospheric carbon levels trap heat and influence Earth's climate.	Group or class discussion during the *Navigate* section.
4. Predict how increasing temperatures might affect processes in Southern California's ocean, both now and in the future.	Class discussion during the *Navigate* section.

Building towards NGSS and AI Literacy:

- **HS-ESS3-6:** Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. In this lesson, students will construct a model based on local climate change and how human activity contributes to positive and negative effects in the environment.

Lesson Overview (Two sessions)

Day 1: Exploring the Greenhouse Effect (50 minutes)

Section	Time	What Happens
Navigate	10 minutes	Students begin the investigation by exploring a graph by the NOAA, showing how California’s average temperature has changed between 1895 and 2025. Students make predictions on why they think this is happening.
Investigate	30 minutes	To investigate this phenomenon, students begin exploring the greenhouse effect by thinking about how a blanket may trap heat and keep someone warm while they sleep. Using this as a foundation, students apply the same concept to Earth’s atmosphere as it comes into contact with light energy from the sun and draw a model to describe this process, setting them up to describe how human emissions of carbon dioxide have impacted this process during day 2.
Navigate	5 minutes	Students share their models as a class, highlighting that light energy that enters the Earth’s atmosphere either leaves the atmosphere as heat energy or bounces back and provides heat on Earth. Students then begin to brainstorm what impacts humans have had on this process through carbon emissions and think about how they can further explore this change.
Problematize	5 minutes	Students reflect on the process of light energy entering the Earth’s atmosphere and think about what they need to explore to further understand how humans have impacted this natural process.

Day 2: Exploring Human-Caused Impacts and Changes to the Greenhouse Effect (50 minutes)

Section	**Time**	**What Happens**
Navigate	10 minutes	Students begin day two by revisiting their greenhouse effect model and thinking about what else they need to explore to describe how humans have impacted this process.
Investigate	30 minutes	During day 2, students work on expanding their model to describe how light energy from the sun interacts with Earth's atmosphere when there are high levels of carbon dioxide present in the atmosphere. Before expanding their model, they explore a Greenhouse Effect simulation by PhET Interactive Simulations. In this simulation, they can manipulate the amount of carbon dioxide in the atmosphere and visually see what happens to light energy as it enters Earth's atmosphere. Following this, they expand their model to show what the interaction between light energy from the sun and Earth's atmosphere looks like when there are increased carbon dioxide levels in the atmosphere. In an optional extension, students watch a video that describes the direct interaction light energy has on carbon dioxide and how it helps in increasing the impact of the greenhouse effect.
Navigate	5 minutes	Students share their models as a class and describe how humans cause carbon dioxide levels in the atmosphere to increase and indirectly impact the Earth's temperature through the greenhouse effect.
Problematize	5 minutes	Student teams reflect on how the greenhouse effect increases the temperature in California and think about how this may also further impact the values that their communities place on Southern California's ocean.

MATERIALS

- Lesson 3.2 Slideshow
- Science Journal
- Whiteboard (1 per group)
- Student PhET Instructions (also in Science Journal, p. 30-33)
- Greenhouse Effect: Waves Simulation
- (optional) How Do Greenhouse Gases Actually Work? [3:08]

BEFORE YOU START TEACHING

1. For **Day 2**, review Student PhET Instructions (also in Science Journal, p. 30-33) to model or support students. Explore Greenhouse Effect: Waves Simulation.

2. Include a numbered list of general steps to set up the investigation before students arrive.

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: In this lesson, we are focusing on helping students understand how human activities influence the greenhouse effect and the resulting changes in Earth's climate. Students are building on their carbon cycle models to explore how rising CO_2 levels trap heat, alter weather patterns, and potentially affect communities and coastal environments in Southern California

Where We Are NOT Going: This lesson does not aim to provide comprehensive predictions of exact climate outcomes or detailed local weather forecasts. We are also not delving into the full complexities of climate modeling, advanced chemistry of greenhouse gases, or policy solutions at this stage.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

DAY 1: NAVIGATE: **California's Average Temperature Over the Years** (10 minutes)

1. As class begins, welcome students to the classroom and remind them that today, they will continue learning about carbon dioxide and its impact on the environment, and ultimately, how changes to carbon dioxide levels will impact the values they place on the ocean.

2. Open the slideshow to **Slide 2** to introduce students to the overview of **Lesson 3.2.**

3. Tell students that for the first task of the lesson, you would like them to work with a partner to make some observations of a graph in the next few slides.

   Over the next few slides, students will make observations on a slow-reveal graph from NOAA, which shows the average temperature from 1895-2025 in the state of California.

4. Advance to **Slide 3**, where students will see the first part of a line graph with no axis or title. Ask students:
   - What do you notice? What do you wonder?
   - How would you describe the trends that you see?

5. After students have had a chance to share, advance to **Slide 4**, and reveal the x-axis (which shows the years 1895-2025) and the y-axis (which shows the average temperature). Ask students:
   - What new information do you have? What else do you wonder?
   - How would you describe the trends that you see?

6. Next, advance to **Slide 5**, which shows the title of the graph. Give students another moment to think and ask them:
   - How would you describe the trends that you see now that you have this new information?

7. Lastly, advance to **Slide 6** to add the 1895-2025 trendline to the graph. Ask students:
   - What do you notice about this new information? What is it telling us?
   - Why do you think we are noticing this trend?

8. Tell students that today, they are going to begin exploring what is causing the trend they are seeing with temperature in California.

DAY 1: INVESTIGATE: **Model** (30 minutes)

1. To begin exploring how the average temperature has changed in California over time, tell students that you will first explore a scenario they can relate to, keeping warm at night.

2. Advance the slideshow to **Slide 7,** where you see an image of a person lying in bed without a blanket. Ask the class the following questions.
    - Where is the heat coming from in this picture?
    - Where does the heat go once it leaves its source?

    As students share their responses, transition the slideshow to display arrows that show heat moving from the heat source (person) into the open air.

3. Then ask students:
    - What would you do if you needed to keep warm?

4. Once students share, reinforce ideas that relate to putting on extra layers, like a blanket. Advance to **Slide 8,** where you see the same image, but this time, with a blanket covering the person. Ask the class the following questions.
    - Where is the heat coming from in this picture?
    - Where does the heat go once it leaves its source?

    As students share responses, listen for responses that you could build upon later, such as the blanket trapping heat.

5. Advance to **Slide 9** and tell students that you would like them to now apply what they have just discussed to a similar process with Earth. To do so, they will draw a model.

6. Hand each student group a blank sheet of paper and advance to **Slide 10,** which shows them how to set up their paper. Optional use: Science Journal, Module 3.2 Student Task: Model and Compare Natural vs. CO2 Atmosphere model, p. 28

7. Transition the slide to show the phrase "**Normal Atmosphere"** appears in the first box. Give students the following task:
    - Draw and label what you think happens to the sun's light energy as it interacts with the normal atmosphere of Earth that hasn't been impacted by other processes.

8. As student groups draw their model, walk around and support their thinking by asking probing and pressing questions.
    - How do you think the sun's light energy interacts with the Earth? How does it interact with the atmosphere?
    - Think back to the blanket model: What purpose did the blanket serve? Is there something similar or different happening with Earth?

9. As students are finishing up their model, gather the group back together to share their findings from their initial model.

DAY 1: NAVIGATE: **Sharing Our Initial Models** (5 minutes)

1. After students have shared their ideas, advance to **Slide 11** and ask students to share the pieces they included in their model.
    - How did the sun's energy interact with the Earth?
    - Where did the sun's energy go once it got to Earth?

   As students share, listen for and reinforce two ideas: (1) light energy enters the atmosphere and leaves as heat energy, and (2) heat energy gets trapped within Earth and heats the planet. You will build upon the second one in the following session, as students learn about elevated CO_2 levels due to human emissions.
   - An example model: https://www.climatecentral.org/graphic/heat-trapping-pollution-2024?graphicSet=Greenhouse+Effect&location=national&lang=en

2. As students share their responses, ask the following questions
    - How do you think humans have impacted this natural process?
    - What have we already learned that can help us expand our model?
    - Where might we learn more about this?

   As students share, listen for connections back to Module 3.1, and humans' impact on carbon dioxide levels in the atmosphere. This is what the following day for the lesson will focus on.

DAY 1: PROBLEMATIZE: **Reflecting on the Day** (5 minutes)

1. As students wrap up for the day, their final task for the day is to reflect on the process of learning about the Earth's atmosphere.

2. Advance to **Slide 12** and have student teams work together to answer the following questions together on **Module 3.2 Day 1 Carbon Dioxide & Weather (Student Reflection), p. 29** in their Science Journals.
    - How might trapped heat and changes in weather patterns impact things your community cares about, like beaches, fishing, or wildlife? Consider both **direct effects** (e.g., warmer ocean water, stronger storms) and **indirect effects** (e.g., changes in recreation, local economy, or ecosystems).

DAY 2: NAVIGATE: **Introduction** (10 minutes)

1. Welcome students back to day two of the CO_2 investigation. Open the slideshow to **Slide 13,** where day two starts.

2. Ask students to get back into their student teams from the last lesson, and when they are ready, advance to **Slide 14**, where they will see an image of the model they began in the previous session.

3. As a class, ask the students the following questions from the last session to get them started.
   - How do you think humans have impacted this natural process?
   - What have we already learned that can help us expand our model?

   As students share their responses, highlight any ideas that connect to humans causing an increase in carbon dioxide emissions that impacts the amount of carbon dioxide in the atmosphere.

4. Move onto **Slide 15**, where the phrase **"Increased CO_2 in Atmosphere"** appears in the empty box of their model.

5. Remind students that in a previous session, they learned about how humans cause an impact on the amount of carbon dioxide in the atmosphere by increasing the amount of carbon dioxide that is emitted.

6. Share with students that the goal of the day is to explore further how this human-caused increase in the levels of carbon dioxide affects how the sun's energy interacts with the Earth's atmosphere.

DAY 2: INVESTIGATE: **Greenhouse Effect** (30 minutes)

1. Advance to **Slide 16** and share PhET Lab, the simulation tool students will use to explore their models further, along with their tasks for the day.
   - As a team, you will investigate PhET's Greenhouse Effect: Waves simulation to explore how light waves are impacted by CO_2 in the atmosphere.
   - On Module 3.2 Day 2: The Greenhouse Effect with PhET Simulations Exploring Wave Interactions (Student Instructions), p. 30-33 of your Science Journal, you will have scenarios to investigate and questions to answer. Work as a team to answer these questions.

2. Once students have opened up their Science Journals and are ready to begin, give them time to explore the Greenhouse Effect: Waves simulation. They will focus on two main scenarios:
   - No carbon dioxide is present in the atmosphere
   - Lots of carbon dioxide are present in the atmosphere

3. As students navigate through the PhET Lab simulation, walk around the group and provide support and ask questions to guide their investigation. Examples include:
    - What do the different color waves represent?
    - When there is no carbon dioxide, what happens to the sunlight when it hits the Earth? What about when you increase the carbon dioxide levels?
    - What happens to the surface temperature as carbon dioxide levels increase? Why do you think this happens?
    - What does the width of the waves represent?

4. If time allows, have students explore the temperature at different time periods (calendar icon). As students explore, ask:
    - What changes did you observe with the land between the ice age, 1750, 1950, and 2020?
    - How did the greenhouse gas levels (carbon dioxide) change during these periods? What impact did it have on the Earth's temperature?

5. Once students have explored the simulation, advance to **Slide 17** and bring them back to their paper models and ask them to draw their observations on the model sheet under the section that reads "**Increased CO_2 in Atmosphere".**

    As students draw their models, walk around and support them by asking probing and pressing questions:
    - Where is the heat coming from in this picture?
    - Where does the heat go once it leaves its source?
    - How do increased levels of carbon dioxide impact this process?

6. As students finish their models, bring the class back together to share their findings.

DAY 2: INVESTIGATE: **Optional Extension**

1. Once students have their models, tell them that you want to share a little more information that can help them refine them even further.

2. Advance to **Slide 18** and play the video How Do Greenhouse Gases Actually Work? [3:08] During this video investigation, students get to explore in depth how carbon dioxide interacts with sunlight to trap heat on Earth.

3. Once students have watched the video, give them a moment to go back to their models and refine them with their newfound knowledge.

DAY 2: NAVIGATE: **Sharing Our Initial Models** (5 minutes)

1. Once students have their models created, advance to **Slide 19** and invite the students to share their models as a class.
    - How does the sun's light energy interact with the Earth in your models?
    - What do increased levels of carbon dioxide have on these interactions?

As students share out loud, reinforce ideas about (1) how increased global temperatures mean that ice caps can melt, raising sea levels in the process, or that (2) excess carbon dioxide can also interact with the ocean water, impacting the ocean's chemistry.

2. Once students have shared, advance to **Slide 20** and ask them to further think about what else they need to explore.
    - How do you think increased CO_2 impacts the value your community places on Southern California's ocean?
    - How might you investigate this?

DAY 2: PROBLEMATIZE: **Reflecting on the Day** (5 minutes)

1. As students wrap up for the day, tell them that their final task is to reflect on what they did.

2. Advance to **Slide 21** and have student teams work together to answer the following questions in **Module 3.2 Day 2: Carbon Dioxide & Weather (Student Reflection)**, p. 34 in their science journal.
    - What did you learn today? How has your thinking changed?
    - How might trapped heat and changes in weather patterns impact things your community cares about, like beaches, fishing, or wildlife?